Denisenko Anna

Big Homework Report

1. Dataset information

For project I`ve chosen Breast Cancer Data Set from UCI Machine Learning Repository.

Number of Instances: 286

This is one of three domains provided by the Oncology Institute that has repeatedly appeared in the machine learning literature.

This data set includes 201 instances of one class and 85 instances of another class. The instances are described by 9 attributes, some of which are linear and some are nominal.

Attribute Information:

1. Class: no-recurrence-events, recurrence-events
2. age: 10-19, 20-29, 30-39, 40-49, 50-59, 60-69, 70-79, 80-89, 90-99.
3. menopause: lt40, ge40, premeno.
4. tumor-size: 0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59.
5. inv-nodes: 0-2, 3-5, 6-8, 9-11, 12-14, 15-17, 18-20, 21-23, 24-26, 27-29, 30-32, 33-35, 36-39.
6. node-caps: yes, no.
7. deg-malig: 1, 2, 3.
8. breast: left, right.
9. breast-quad: left-up, left-low, right-up, right-low, central.
10. irradiat: yes, no.

In dataset they all are represented as categorical.

I have to mention that our data is unbalanced (201 against 85 examples) and this may cause some problems with classification.

I cleaned it from missing data and later separated for 10 train — test sets.

2. Binarization procedure

As all our data is categorical, we just separate our colums into several (for example, «breast: left», «breast: right») and place 1 for column name that our object has and 0 otherwise.

3. Classification procedure

Two algorithms worked for me:

1. Voting based on fact of intersection: if example`s not empty with some context, we give him a point. The winner is one who has more points.

2. Voting that takes length of intersection into consideration. Bigger intersection get more points(depends on it`s length). Again, the winner is one who has more points.

First algorithm got accuracy = 0.785, second got accuracy = 0.821, but accuracy depends on train-test set I choose. I think it“s because of inbalance of data. Also when I tried some algorithms with checking counter — examples, they were always found, so it didn`t help me to classify objects.